

AMENDMENT NO. 1 JUNE 2013
TO
IS 1528 (Part 1) : 2010 METHODS OF SAMPLING AND PHYSICAL TESTS FOR
REFRACTORY MATERIALS
PART 1 DETERMINATION OF PYROMETRIC CONE EQUIVALENT (PCE) OR
SOFTENING POINT

(Third Revision)

(Page 1, clause 4.3.3) — Insert the following at the end of clause:

‘Subsequently make specimen as specified in 4.4.1’.

(Page 1, clause 4.3.4) — Substitute the following for the existing clause:

‘4.3.4 For Mortars

Make specimen as per dry monolithics materials and add the liquid component, if any, in appropriate proportion. Subsequently make specimen as specified in 4.4.1’.

(Page 2, clause 4.4.1, line 1) — Substitute ‘4.3’ for ‘4.3.2’.

(Page 2, clause 4.4.1, line 5) — Substitute ‘28 mm’ for ‘25 mm’.

(Page 2, clause 4.5.2, line 7) — Substitute ‘(see Fig. 2 as an example of orton cone)’ for ‘(see Fig. 2)’.

(Page 2, clause 4.5.2) — Insert the following at the end:

‘The test pieces and the pyrometric reference cones shall be positioned so that an edge or a face leans inward at the Angle of 8° to the vertical’.

(Page 2, clause 4.5.5, line 1) — Insert ‘the mean rate’ for ‘the rate’.

(Page 2, Fig. 1) — Substitute the following figure for the existing figure:

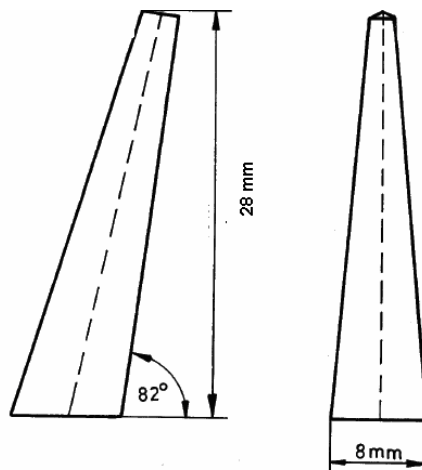


FIG. 1 STANDARD PYROMETRIC CONE

(Page 3, Table 1) — Substitute the following for the existing table:

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**Table 1 Summary of Reference Temperatures and Cone Designations of
ISO, Segar, Orton and HPC Pyrometric Reference Cones**

Reference Temperature °C (1)	IS O (2)	Segar (3)	Orto n (4)	HPC (5)	Reference Temperature °C (1)	ISO (2)	Segar (3)	Orton (4)	HPC (5)
605	-	022	-	022	1 125	158	01a	-	2a
650	-	021	-	021	1 145	160	1a	-	3a
675	-	020	12	020	1 165	-	2a	23	4a
695	-	019	13	019	1 185	162	3a	26	5a
715	-	018	-	018	1 200	-	-	-	6a
735	-	017	-	017	1 220	-	4a	-	-
760	-	016	14	016	-	-	-	-	-
785	-	015a	-	015a	1 230	164	5a	27	7
815	-	014a	-	014a	1 250	-	-	-	8
835	-	-	-	013a	1 260	-	6a	28	-
845	-	013a	15	-	1 270	-	7	-	-
855	-	-	-	012a	1 280	-	-	-	9
880	-	-	-	011a	1 295	166	8	29	10
890	-	012a	-	-	1 315	-	9	-	11
-	-	-	-	-	1 330	-	10	-	-
900	-	011a	-	010a	1 340	-	-	12	-
-	-	-	-	-	1 350	-	11	13	12
925	-	010a	16	09a	-	-	-	-	-
940	150	09a	-	08a	1 375	-	12	-	13
-	-	-	-	-	1 395	-	13	14	-
965	-	08a	17	07a	-	-	-	-	-
975	152	07a	18	06a	1 410	-	14	-	14
-	-	-	-	-	-	-	-	-	-
995	-	06a	-	05a	1 430	-	-	15	15
1 010	154	05a	19	-	-	-	-	-	-
1 020	-	-	-	04a	-	-	-	-	-
1 040	-	-	-	03a	1 440	-	15	-	-
1 055	-	04a	-	02a	1 460	-	-	-	16
1 070	156	03a	20	-	1 470	-	16	-	-
1 080	-	-	-	01a	1 480	-	-	-	17
1 100	-	02a	-	1a	1 490	-	-	16	-
-	-	-	-	-	1 500	150	17	-	18

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Reference Temperature °C (1)	IS O (2)	Segar (3)	Orton (4)	HPC (5)
-	-	-	-	-
1 510	-	-	17	-
1 520	152	18	18	19
-	-	-	-	-
1 530	-	-	-	20
1 540	154	19	19	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
1 560	156	20	20	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
1 580	158	26	-	26
1 600	160	27	-	-
1 605	-	-	23	27
1 620	162	27½	26	-
-	-	-	-	-
-	-	-	-	-
1 630	-	-	-	28
1 640	164	28	27	-
-	-	-	-	-
1 646	-	-	28	29
-	-	-	-	-
-	-	-	-	-
1 660	166	29	29	-
1 665	-	-	30	30
1 680	168	30	31	-
1 690	-	-	-	31
-	-	-	-	-
1 700	170	31	31½	-
1 710	-	32	-	32
-	-	-	-	-
1 720	172	32½	32½	-

Reference Temperature °C (1)	ISO (2)	Segar (3)	Orton (4)	HPC (5)
-	-	-	-	-
-	-	-	-	-
1 730	-	33	-	33
-	-	-	-	-
-	-	-	-	-
1 740	174	33½	33	-
-	-	-	-	-
-	-	-	-	-
1 750	-	-	-	34
1 760	176	34	34	-
-	-	-	-	-
1 770	-	-	-	35
1 780	178	35	-	-
1 785	-	-	35	36
1 800	180	36	36	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
1 820	-	-	37	37
1 830	-	37	-	-
-	-	-	-	-
1 835	-	-	38	-
1 850	-	-	-	38
1 860	-	38	39	-
-	-	-	-	-
1 880	-	39	40	39
-	-	-	-	-
-	-	-	-	-
1 900	-	40	-	-
1 920	-	-	-	40
1 940	-	41	-	-
1 960	-	-	-	41
1 970	-	-	41	-

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Reference Temperature °C (1)	ISO (2)	Segar (3)	Orton (4)	HPC (5)
1 980	-	42	-	-
2 000	-	-	-	42
2 015	-	-	42	-

NOTE — Any standard cone is acceptable.
